

# KESSLER

## Quality Assurance Report — Release Readiness

**Document ID:** KSL-QA-001      **Version:** 1.0      **Date:** 21 July 2026

**Verdict:** READY FOR LOCALHOST TESTING

**Baseline:** KSL-SRS-001 v1.0 · commit ba7b31d+

## 1. Summary

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KESSLER v1.0 was verified against the SRS acceptance criteria. The automated suite (**45 tests, 45 passed, 0 failed**, 0.82 s) covers the Pc mathematical core against closed-form references, CDM parsing (KVN + cdm\_public), screening on engineered geometries, risk-engine thresholds, and the full API contract. Live end-to-end verification was performed on localhost (backend :8000, UI dev server). One environment deviation and two minor findings are recorded below; none block testing. **Verdict: the application is ready to be tested.**

## 2. Automated Test Results

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| Suite                           | Coverage                                                                                                                                                                                                        | Result |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| test_collision_probability (13) | Foster vs isotropic closed form ( $<1e-3$ rel); Foster/Chan agreement on 3 geometries; Max-Pc upper-bound property across $\sigma$ sweep; boundary behaviours; dispatch paths (computed/reported/max)           | PASS   |
| test_cdm_parser (6)             | KVN round-trip incl. RTN covariance symmetry; aggregated validation errors; non-PSD covariance quarantine; cdm_public row parsing incl. km $\rightarrow$ m and null-PC                                          | PASS   |
| test_screening (5)              | Engineered close pair detected ( $<25$ km, RTN components present); altitude sieve excludes disjoint orbits; empty-catalogue no-op                                                                              | PASS   |
| test_risk_engine (9)            | Threshold boundaries at $1e-4/1e-5/1e-7$ ; sub-km covariance-free escalation (capped at CRITICAL); urgency monotonicity in class/time/staleness; bounds                                                         | PASS   |
| test_api (12)                   | Health; urgency-sorted list w/ attribution; detail + X-Request-Id; 404/422 error envelopes; insight fallback; screening incl. window-limit rejection; pipeline status shape; ISS track sanity; catalogue search | PASS   |

### 3. Live Verification (localhost)

| Check                                      | Observed                                                                                                                       | Status |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|
| Backend boot (Demo Mode, zero credentials) | Pipeline ran on startup; 8/8 demo CDMs published; all 8 agents report <code>ok</code>                                          | ✓      |
| GET /api/v1/conjunctions                   | 8 events, urgency-descending; CRITICAL events top-ranked; attribution present                                                  | ✓      |
| Max-Pc fallback path                       | Events with null PC correctly labelled <code>pc_type: "max"</code> and escalated per FR-RSK-1                                  | ✓      |
| GET /satellites/25544/track                | 61 points; ISS altitude 413.5 km (physically correct); ECI vectors present for 3D view                                         | ✓      |
| POST /screening/run (demo asset 61001)     | Altitude sieve reduced 10-object catalogue to 2 candidates; engineered conjunction pair found at 23.6 km                       | ✓      |
| Error envelope + observability             | 404 returns <code>NOT_FOUND</code> envelope; request-id propagated to logs and X-Request-Id header; rotating file log active   | ✓      |
| UI production build                        | <code>tsc</code> strict + Vite build clean (628 modules)                                                                       | ✓      |
| UI dev server + API proxy + diagram        | Page serves with KESSLER title; /api proxy returns backend JSON; n8n-style diagram served and postMessage status channel wired | ✓      |
| AI insight graceful degradation            | No API key configured → deterministic template briefing returned, <code>source: "template"</code> , no error                   | ✓      |

### 4. Findings

| ID   | Severity | Finding                                                                                                                                                                                                                                                              | Disposition                                       |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| QA-1 | Note     | <b>Runtime deviation:</b> host machine offers Python 3.9.7 (silent 3.12 installs are blocked on this box). Backend runs on 3.9 via <code>eval-type-backport</code> (declared conditionally in requirements). All 45 tests pass on 3.9; target runtime remains 3.11+. | Accepted; revisit when Python ≥3.11 is installed. |
| QA-2 | Minor    | Port 5173 may be occupied by another Vite project on this machine; KESSLER auto-falls back to 5174. CORS allowlist updated to cover both.                                                                                                                            | Fixed in this release.                            |
| QA-3 | Minor    | UI bundle is ~1 MB minified (three.js). Acceptable for localhost; code-splitting deferred to the deployment phase.                                                                                                                                                   | Deferred (tracked for AWS phase).                 |
| QA-4 | Info     | Demo dataset TCAs are fixed calendar dates; countdowns will read "TCA passed" as the demo ages. Live mode is unaffected.                                                                                                                                             | By design for a static demo dataset.              |

## 5. Requirements Traceability (Must-haves)

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FR-ING-1..5 ✓ (demo fallback, TTL cache, last-known-good) · FR-CDM-1..4 ✓ · FR-PC-1..5 ✓ (Foster + Chan cross-check + Max-Pc, pure logic layer) · FR-SCR-1..4 ✓ (FR-SCR-5 performance target met on demo catalogue; full-catalogue benchmark deferred to live-mode testing) · FR-RSK-1..3 ✓, FR-RSK-4 ✓ (template fallback verified) · FR-API-1..3 ✓ · FR-UI-1..5 ✓ · NFR-1/2/4/5/7 ✓ · NFR-3 spot-checked (cached list <50 ms local) · NFR-6 partially measured (logic-layer suites comprehensive; coverage tooling not yet wired).

## 6. Sign-off

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QA verdict: **APPROVED for localhost testing**. Recommended next steps: user acceptance testing on the dashboard, Space-Track credentials for live-mode soak, then the AWS deployment phase per HLD §6.